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Paleomagnetism of the Chiapas Massif, southern Mexico: Evidence for rotation of the Maya Block and implications for the opening of the Gulf of Mexico

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ABSTRACT

We have analyzed 180 oriented samples of the batholithic complex and overlying red beds of the Chiapas Massif in southern Mexico for a paleomagnetic study. Both stepwise alternating-field and thermal demagnetization procedures revealed dual-polarity characteristic magnetizations above 10–30 mT or 200–400 °C in about 60% of the samples demagnetized. Three distinct and ancient paleomagnetic directions have been identified. Late Permian intrusions (256 ± 10 Ma) yield nearly equatorial southern hemisphere paleolatitudes and a paleopole (13.4°S, 176.1°E; 12 sites, K = 15.4, A_{95} = 11.4°) rotated ~75°, counterclockwise, with respect to the North American reference pole. A selected set of Late Triassic(?)–Jurassic red beds of the Todos Santos Formation yields a paleopole at 22.6°N, 170.6°E (4 sites, K = 56.8, A_{95} = 12.3°), which implies ~60° counterclockwise rotation, and a near equatorial northern hemisphere paleolatitude. Middle-Late Jurassic dikes and granites yield a paleopole at 77.4°N, 97.4°E (6 sites, K = 269.2, A_{95} = 4.1°) near the Mid-Jurassic segment of the North American apparent polar wander path. These data are consistent with a pre-rift configuration in which the Maya Block is rotated to a position off the coast of Texas and Louisiana in the present-day area of the Gulf of Mexico. A further implication is that by Oxfordian times, Chiapas (and by inference the Maya Block) was essentially in its current position with respect to cratonic North America.

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INTRODUCTION

A problematic aspect of Late Paleozoic–Early Mesozoic reconstructions of the western part of equatorial Pangea is the overlap of South America with southern Mexico and Central America (Bullard and others, 1965; Freeland and Dietz, 1972; Van der Voo and others, 1976; Pilger, 1978; Walper, 1980; Pindell, 1982, 1985; Klitgord and Schouten, 1986). This implies that Precambrian and Paleozoic terranes in southern Mexico are "suspect" in the sense that they are thought to have arrived at their present position in the North American plate in post-Pangean times. The allochthonous nature of terranes in southern Mexico has recently been recognized on the grounds of geologic and geochemical evidence (Campa and Coney, 1983; Ortega-Gutierrez, 1981; Carfantan, 1983; Ruiz and others, 1987). Furthermore, paleomagnetic techniques have provided support for this interpretation (Gose and Sanchez-Barreda, 1981; Ballard and others, 1989; Fang and others, 1989; McCabe and others, 1988). Paleomagnetic data for this area, however, are insufficient, and Paleozoic units have been affected by remagnetizations (for example, McCabe and others, 1988). Thus, the spatial relationships of terranes in southern Mexico with respect to the larger continental masses in the late Paleozoic are poorly constrained. The objective of this investigation is to provide paleomagnetic data from the Maya (or Yucatan) Block, in southern Mexico, in order to constrain its possible location in reconstructions of Pangea and evaluate those models previously proposed for the relative rotation and displacement of the Maya Block during the tectonic evolution of the Gulf region.

In recent evolutionary models of the Gulf/Caribbean region, it has been suggested that the Maya Block originated in the present area of the Gulf of Mexico and that it separated from stable North America, forming the Gulf of Mexico in the Jurassic. There is, however, some disagreement on the actual timing and motion mechanism. Thus far, it has been proposed that motion of the Maya Block was synchronous with a period of sea-floor spreading in Middle-Late Jurassic times. Motion could have been accommodated by lateral displacements along large faults, and some of the structures postulated are as follows: the Salina Cruz Fault (Viniegra, 1971), the Tamaulipas-Chiapas Fault (Pindell, 1985), a structure parallel to the eastern extension of the Sonora-Mojave megashear (Klitgord and Schouten, 1986), or the proto–Polochic-Motagua structure (Anderson and Schmidt, 1983). Accordingly, models for the actual motion of the Maya Block result in different estimates of the total amount and sense of tectonic rotation since it separated from North America. A large anticlockwise rotation of the Maya Block is required, for instance, in the model of Carey (1963), and more recently, Ross and Scotese (1988), Dunbar and Sawyer (1987), Pindell (1985), and Anderson and Schmidt (1983). On the other hand, clockwise rotation has been proposed by Freeland and Dietz (1972), whereas Salvador (1987), Klitgord and Schouten (1986), and Dickinson and Coney (1980), among others, show the Maya Block close to its present orientation relative to North America. The latter models suggest that the amount of rotation of the Maya Block, if any, would have been small. In the simplest kinematic models, the Maya Block and other small continental blocks are attached to one of the larger continental masses, whereas more complicated models imply independent microplate motions.

Previous paleomagnetic investigations of red

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beds of the San Ricardo Formation in central Chiapas (Guerrero and others, 1990) indicate that discernible rotation of the Maya Block, with respect to the North American craton, has not occurred since the latest Jurassic. On the other hand, paleomagnetic directions for limestones of the lower Permian Paso Hondo and Grupera Formations in southernmost Chiapas (Gose and Sanchez-Barreda, 1981) yield an equatorial paleolatitude for the sampling area and a paleopole that plots near the Jurassic segment of the North American apparent polar wander path (APWP). The age of the magnetization is not constrained by a field test, yet the data have been interpreted to indicate 22° clockwise rotation of the Maya Block. In this paper, we report new paleomagnetic results from the Chiapas Massif in the southwestern portion of the Maya Block. The paleogeographic and tectonic implications of the results are discussed in the context of the models proposed for the rotation of the Maya Block and the opening of the Gulf of Mexico in Mesozoic times.

The Maya Block (Fig. 1; Campa and Coney, 1983) comprises the Yucatan peninsula, the southern portion of the Gulf coastal plain, the adjacent continental shelf, and the base of the peninsula to the Pacific Ocean north of the Polochic-Motagua suture zone (Chiapas, Belize, and portions of Guatemala). The tectonic coherence of Yucatan and Chiapas in the Late Permian (and at later times) cannot be demonstrated. Following Pindell's (1985) suggestion, some recent tectonic models have excluded the Chiapas Massif from the Maya Block, but for reasons that are unclear. The eastern outcrops of the Permian batholith are disconformably overlain by continental and marine formations of Mesozoic age. Similarly, drill-hole data indicate that Permo-Triassic granites form the basement of the Mesozoic platformal and basinal sequences to the north and east in the Gulf of

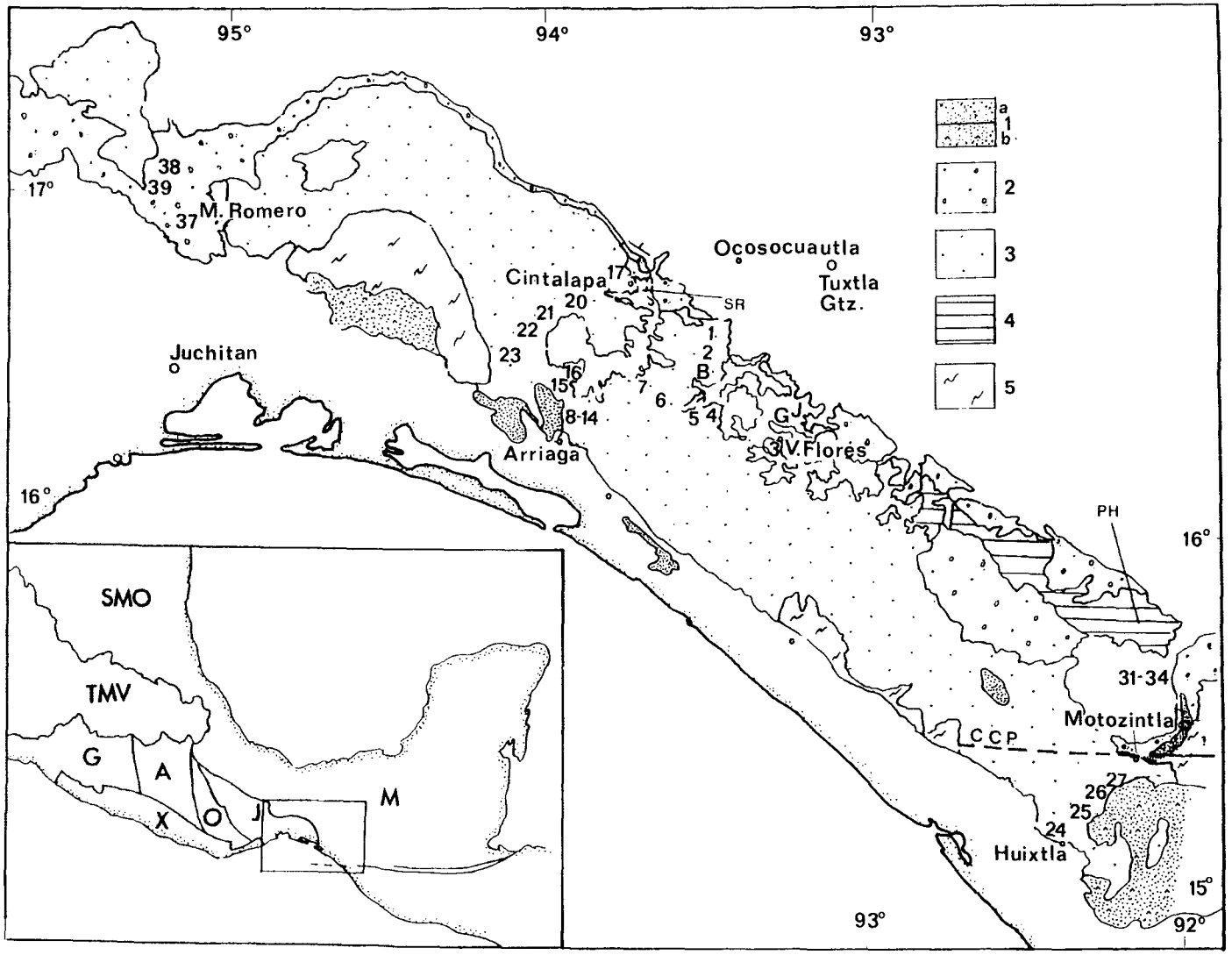


Figure 1. Simplified geologic map of western Chiapas, Mexico. The numbers indicate paleomagnetic sampling sites. SR and PH indicate the localities sampled by Guerrero and others (1990) and Gose and Sanchez-Barreda (1981) in the San Ricardo and Paso Hondo-Grupera Formations, respectively. The dashed line follows the trend of the Cuilco-Chixoy-Polochic (CCP) fault system. (1) Cenozoic intrusions (a) and volcanics (b); (2) Todos Santos Formation; (3) Chiapas batholith; (4) Pennsylvanian and Lower Permian Paso Hondo and Grupera Formations; (5) undifferentiated metamorphics. Inset: Terrane map of southern Mexico. G = Guerrero; A = Acatlan; O = Oaxaca; X = Xolapa; J = Juarez; SMO = Sierra Madre Oriental; TMV = Trans-Mexican Volcanic Belt; M = Maya Block (Yucatan).

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Mexico coastal plain (Lopez-Ramos, 1979). In fact, there is no evidence suggesting that the eastern margin of the Chiapas Massif is a crustal discontinuity or a suture. Although faulting has been observed in the southeastern portion of the Chiapas Massif, the faults are Cenozoic structures to the east of outcrops of the upper Paleozoic Santa Rosa and Grupera Formations. These units are intruded by the Permian batholith of Chiapas, and they have been correlated with the late Paleozoic sequence of the Maya Mountains in Belize (Hall and Bateson, 1972). For these reasons, we have made the assumption that results from Chiapas may be extrapolated to the entire Maya Block.

GEOLOGICAL SETTING AND SAMPLING

In the southwestern portion of the Maya Block, in western Chiapas, metamorphic rocks of uncertain age are intruded by Late Permian granites and granodiorites (10-point, whole-rock, Rb-Sr isochron with an apparent age of 256 ± 10 Ma by Damon and others [1981], and multiple K-Ar determinations that range between 219 Ma and 246 Ma). The Permian intrusives form a batholithic complex elongated in the northwest-southeast direction, which appears to extend to the northwest, across the Tehuantepec isthmus, into the Mixtequita batholith. Near Motozintla, in southern Chiapas, the Permian batholith intrudes the Santa Rosa Formation, which contains an Early Pennsylvanian fauna of crinoids (Buitron, 1977) and a metamorphic complex that has yielded a latest Proterozoic age (Rb-Sr isochron of 606 Ma for an amphibolite gneiss, P. Damon, 1989, personal commun.).

The Todos Santos Formation (Sapper, 1899) is a continental sequence of red beds, conglomerates, and minor volcanics that overlies the Permian granites and underlies Lower Cretaceous or Upper Jurassic marine units. The age of the Todos Santos Formation is not well established because it generally lacks diagnostic age fossils. The lower age limit is more uncertain, and it could be as old as Late Triassic (Mullerried, 1942; Lopez-Ramos, 1979), but the minimum age is clearly pre–latest Jurassic. The Todos Santos Formation, near Cintalapa in central Chiapas, is disconformably overlain by the San Ricardo Formation, which contains a Kimmeridgian-Portlandian fauna (Alencaster, 1977). Moreover, in the northern portion of the Tehuantepec isthmus, the Todos Santos Formation is overlain by the Late Jurassic Mogone Formation, which is dated by the presence of *Rhaxella sorbayana* (Carfantan, 1981). In Guatemala, north of the Polochic fault, Richards (1963) has suggested that the age of the Todos Santos Formation is restricted to the Late Jurassic, on the basis of the occurrence of diagnostic Late Jurassic fossils in a basal conglomerate. In the subsurface of Belize and Yucatan, a relatively thin section of red beds that overlies basement rocks and underlies the Cretaceous section has been tentatively correlated with the Todos Santos Formation (Lopez-Ramos, 1979).

A Middle to Late Jurassic (174–141 Ma) magmatic event in Chiapas has been proposed by Damon and others (1981) based on reset K-Ar biotite ages of Permian granites, and radiometric dating of andesitic dikes and hydrothermal alteration products. Jurassic granites have not been separately mapped in published geologic maps.

For the present study, we sampled primarily the Late Permian intrusives of the Chiapas batholith. The overlying red beds of the Todos Santos Formation and basaltic and andesitic dikes that intrude the granites were sampled in less detail. The locations of sampling sites are shown on a simplified geologic map of western Chiapas in Figure 1. The batholithic rocks collected range in composition from granite to diorite. Although an effort was made to collect only the less-weathered material, we were limited to outcrops existing in, or near, passable roads. Different facies of the Permian batholith were sampled along the highways that connect the cities of Tepanatepec, Arriaga, Villa-Flores, Cintalapa, and Ocosocuautla. Farther south along the Huixtla-Motozintla highway (Fig. 1) we sampled batholithic sites south of the presumed western extension of the Cuilco-Chixoy-Polochic (CCP) fault system.

The Todos Santos Formation was sampled in three areas, and stratigraphic intervals of about 5 m were sampled at each site. We collected four sites near Motozintla (Fig. 1) north of the Amatenango fault (the north splay of the CCP fault system), three sites more in the region of Tehuantepec, near Matias Romero, and, finally, we obtained two sites from a volcanic unit of Jurassic age near Cintalapa (Meneses-Rocha, 1985). At this last locality, we also collected granite clasts from a cross-bedded conglomeratic sandstone of the upper portion of the Todos Santos Formation for a conglomerate test.

In the isthmus of Tehuantepec, the Todos Santos Formation is locally folded and is seen to overlie granitic basement (La Mixtequita granite). Deformation in this area is associated with the accretion of an island-arc assemblage (the Chontal arc) during the Early Cretaceous (Carfantan, 1983). Near Cintalapa, the Todos Santos Formation is relatively undisturbed; it overlies Permian granites and dips gently to the northeast. Greater structural complexity exists in the Motozintla region. There, the sites collected in the Todos Santos lie east of fault-bounded blocks of granite, low-grade metamorphic volcano-plutonic rocks, and amphibolite gneisses, exposed along the trace of the CCP fault system.

A total of 37 sites (28 on intrusive rocks) were collected; samples from nearly contiguous sites (1 and 2; 24 and 25) were combined yielding a total of 35 sites. At 8 sites, either the magnetic behavior was unstable, or only paleomagnetic directions clearly similar to the late Tertiary or the present-day field were observed. Thus, our interpretations are based on the analysis of the remaining 27 sites (134 samples). The samples were collected as both hand samples and drilled cores. All samples were oriented with a magnetic compass. Orientations were determined at different distances from the outcrop and in several directions, showing that effects of the outcrop magnetization on the compass reading are negligible. In the laboratory, one to three specimens were prepared from each sample.

PALEOMAGNETIC ANALYSIS

Natural remanent magnetizations (NRMs) were measured with either a two-axis ScT cryogenic magnetometer or a Schonstedt spinner magnetometer at the paleomagnetic laboratory of the University of Michigan. All samples were subjected to stepwise thermal or alternating-field (AF) demagnetization. A few samples from the Todos Santos Formation were also chemically demagnetized. Results from the intrusives and the red beds will be discussed in separate sections. Magnetic components were identified in orthogonal demagnetization diagrams (Zijderveld, 1967). Sample directions and/or great circle trajectories were calculated using principal component analysis (Kirschvink, 1980). Site mean directions and formation means were calculated using Fisher statistics (Fisher, 1953).

Intrusives

NRM directions for the intrusive rocks are plotted on an equal area projection in Figure 2a (including the directions of the sites rejected). Directions are predominantly to the north and downward, indicating the presence of strong overprints of late Tertiary or more recent age. Other samples form scattered groups to the east and west at shallow to intermediate inclinations. NRM intensities are relatively high. The mean site intensities of the NRM range between 6.14 and 650 mA/m, except at sites 24 and 25, where the maximum intensities reach 6 A/m.

Progressive demagnetization diagrams are shown in Figures 3 and 4. In both figures, de-

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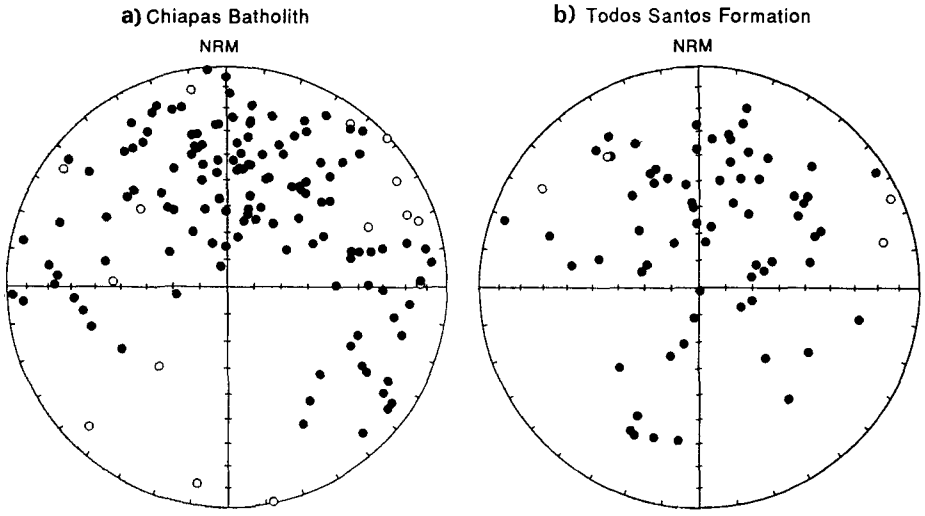


Figure 2. Plot of the natural remanent magnetization (NRM) of the specimens collected on equal area projections (geographic coordinates). (a) Chiapas batholith; (b) Todos Santos Formation.

magnetization plots indicate that the majority of the samples exhibited one or two components of magnetization. A northerly and downward overprint (component A) is easily removed during thermal treatment as high as 200 to 400 °C or inductions of about 10–30 mT (see Figs. 3a, 3e, 3f, 4a, and 4b). The remaining magnetization unblocks at temperatures of 540 to 590 °C (or relatively high inductions sometime >100 mT).

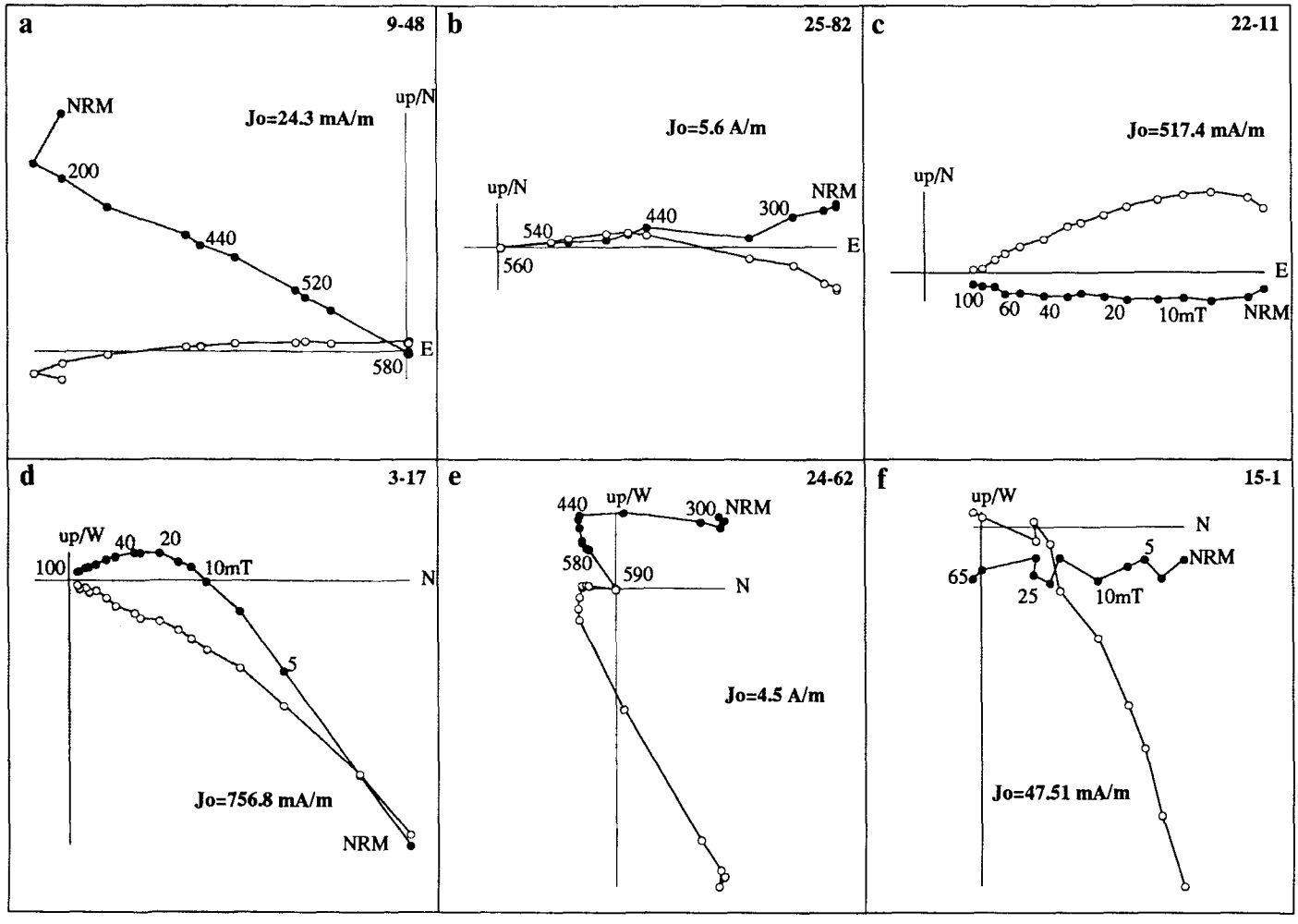


Figure 3. (a–f) Representative demagnetization diagrams (Zijderveld, 1967) of the Permian intrusives of the Chiapas batholith. Open (closed) symbols indicate projections in the vertical (horizontal) plane. Inductions are indicated in mT (c, d, and f) and temperatures in °C (a, b, and e). Jo = initial magnetization.

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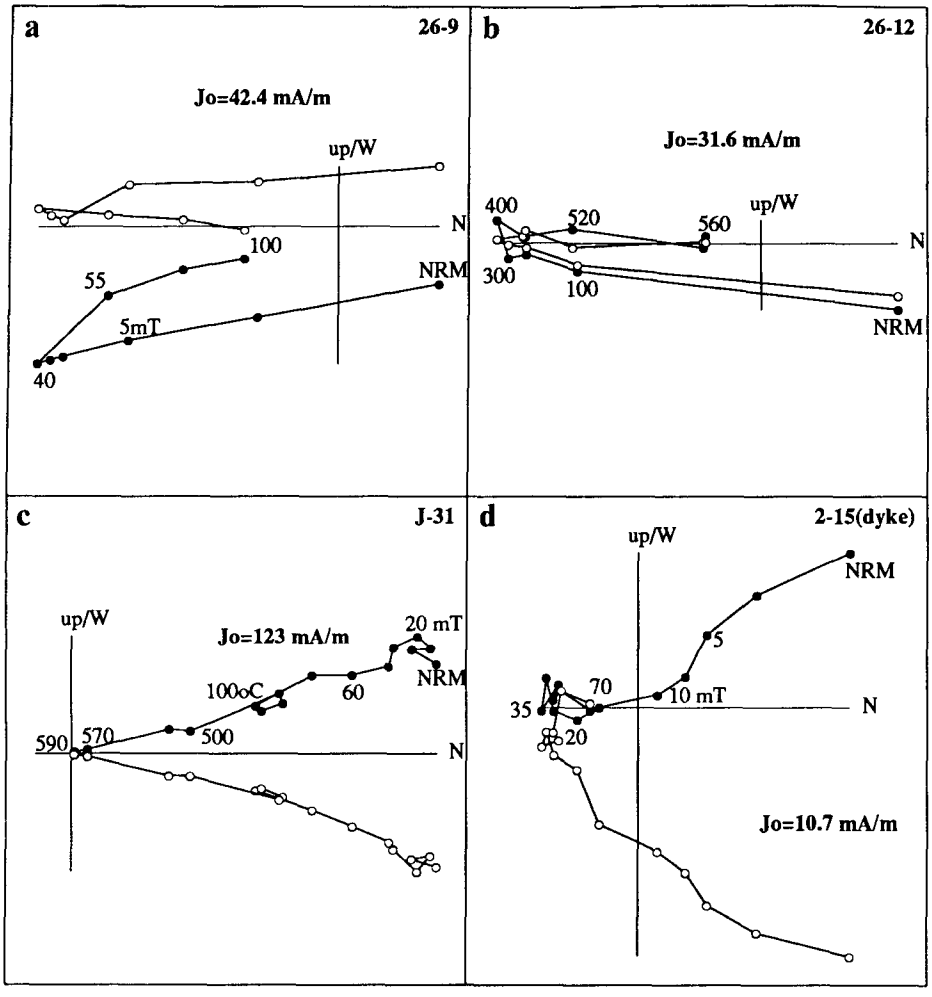


Figure 4. (a–d) Representative demagnetization diagrams showing specimens from sites where the characteristic magnetization has been interpreted as a Middle-Late Jurassic TRM or pTRM (component B). Symbols are as in Figure 3. The sample from Figure 4d was obtained from an undated andesitic dike that intrudes the Permian batholith. In this sample, the characteristic magnetization is to the south and shallow, while in the hosting granite the characteristic magnetization is westerly and shallow. This suggests that the B component is relatively younger than the C component of Figure 3.

This high temperature (coercivity) component is considered the characteristic remanent magnetization (ChRM). In most cases, consecutive colinear points that trend to the origin in demagnetization diagrams indicate the complete isolation of the ChRM, free of any overprint (for example, Figs. 3a, 3b, 3c, and 4c). In a few specimens, however, the trajectories of demagnetization plots fail to reach the origin, or they form curved segments, which indicate the unsuccessful isolation of a single magnetic component due to greater overlap in the coercivity or unblocking temperature spectra of the A component and the ChRM (see Figs. 3d and 3f). The site mean of sites with this kind of behavior was calculated using the technique of Bailey and Halls (1984).

Two groups of sites with distinct characteristic directions can be recognized in the intrusives. The prevalent directions are easterly (or westerly) and relatively shallow (Fig. 3), but in several sites (6, 23, J, 26, 27), the ChRM that is isolated is either to the north or south, with nearly horizontal inclination (Fig. 4). In both cases, the maximum unblocking temperatures and coercivities are typical of (titano) magnetite or maghemite. We refer to the easterly (westerly) component of Figure 3 as the C component, whereas we will refer to the northerly (southerly) component of Figure 4 as component B. An undated andesitic dike sampled in site 2 carries a southerly and shallow magnetization (component B), that is revealed after AF treatment of 30 mT (Fig. 4d). The B direction of the dike is different from the westerly (C) magnetization of the hosting granite. Thus, these components are clearly of different age, and it appears that the B component is of a younger age than the C component. The specimen directions and poles to great circle trajectories for the east-west C component and the north-south B component are plotted in Figures 5a and 5b, respectively. Site mean directions and associated statistical parameters are summarized in Table 1.

Todos Santos Red Beds

NRM directions for the sites collected in the Todos Santos Formation are plotted in Figure 2b. NRM directions are considerably scattered. Typical demagnetization diagrams are shown in Figures 6 and 7. Figures 6a and 6b show the behavior for the great majority of the samples collected near Matias Romero. The samples are characterized by two components of magnetization. The first component removed generally unblocks at temperatures that range between 100 and 450 °C; this component is similar in direction and stability to component A discussed above. The second magnetic component removed unblocks at temperatures typical of hematite and has a (tilt-corrected) dual polarity magnetization with west-northwest–east-southeast declinations and shallow inclinations. As seen in Figures 6a and 6b, this component is well defined by linear segments that trend to the origin. We consider this component the characteristic remanent magnetization of the Todos Santos Formation. A similar component of magnetization can be identified in the trend of the magnetization vector during chemical demagnetization experiments, but only after treating the sample for 50 hr (Fig. 6c). This is due to the overlap in the solubility spectra of the mineral grains carrying the steeper overprint and the ChRM. In four specimens from two different sites, in addition to the northerly and the easterly components, a third component of magnetization was observed with still higher unblocking temperatures. This component of magnetization has southerly declinations and nearly horizontal inclinations, similar to the B component observed in dikes and some of the granites (for example, sample 37-41, Fig. 6d). The southerly directions do not change significantly upon structural correction. As is the case for the intrusions, these two components are clearly of different age, but here their relative time of acquisition cannot be directly determined. The specimens illustrated in Figures 6e and 6f are from the volcanic unit sampled near Cintalapa. Of the two sites collected, only samples from site 17 yielded stable magnetizations. The samples are characterized by high intensities and rela-

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TABLE 1. PALEOMAGNETIC DATA AND STATISTICAL PARAMETERS

Site	No/N/n			dec (°)	inc (°)	k	α_{95}(°)	VGP	
								°E	°N
Permian intrusives C component									
1,2	14/9/11*			243.3	−11.3	14.2	21.5	179.6	−27.2
3	7/4/ 5*			264.5	−4.6	15.3	24.8	177.1	−5.9
5	7/3/ 3			233.3	−12.1	58.5	16.3	182.2	−36.9
8	10/4/ 4			253.2	2.6	12.9	26.6	183.2	−15.7
9	5/4/ 4			297.4	−22.8	31.2	16.7	156.7	22
10	5/4/ 5			48.5	34.0	94.3	9.5	166.6	−43.8
14	4/3/ 3			245.0	16.1	71.0	14.7	192.9	−21.2
15	5/8/ 8*			280.4	−1.2	28.0	12.4	173.6	9.8
16	4/4/ 4			79.7	28.3	37.0	15.3	170.7	−12.5
20†	2/1/ 2			241.5	18.5			195.9	−23.8
21†	2/1/ 2			221.0	−36.0			165.6	−51
22	5/4/ 5			89.5	−4.7	8.2	28.4	179.4	0.2
24, 25	5/5/10			87.7	7.0	10.6	16.6	175.7	−3.1
B	4/3/ 3			68.1	19.9	110.0	11.8	173.6	−23.7
Mean	12			261.3	−9.0	10.0	15.0	176.1	−13.4
T.C.				263.0	2.5	10.0	15.0	179.0	−9.4
B component									
2	2/2/ 2			176.1	−3.9			102.6	74.9
6	5/3/ 5			4.4	10.0	71.0	14.7	66.3	77.8
23	4/4/ 8*			170.5	−6.5	15.7	21.6	123.3	73.8
26	4/4/ 6			168.3	−5.3	35.3	15.7	96.1	78.6
27	3/3/ 9			178.5	−7.6	28.8	23.4	105.2	77.8
J	6/6/ 6			1.5	8.3	109.8	6.4	80.7	77.8
Mean	6			176.5	−7.0	153.0	5.4	97.4	77.4
T.C.				177.2	0.9	153.0	5.4	97.6	73.3
Todos Santos Formation									
		dec (°)	inc (°)	dec (°)	inc (°)				
(a) Matias Romero–Cintalapa									
17	5/4/ 7	295.1	−4.5	296.6	−6.8	53.0	10.6	166.1	22.9
37	5/5/11*	283.2	0.6	285.5	15.9	9.6	18.5	179.5	17.1
38	3/3/ 3*	309.1	50.0	286.0	−1.0	11.0	20.2	170.7	15.1
39	3/3/ 6	127.2	10.8	126.3	−2.8	14.6	23.7	165.4	35.0
Mean	4	297.6	14.2	293.7	2.7			170.6	22.6
		k=9; α95=38°		k=34; α95=15.8°					
McFadden outlier test: γ=49.1°		p=0.141	γ=15.6°	p=0.965					
		78% unf. (max k)		295.7	4.6	k = 36.1			
(b) Motozintla		*in situ*		(tilt corrected)					
31	4/4/ 9	174.0	12.0	170.5	0.8	6.8	31.7	118.7	71.2
32	5/4/10	199.7	17.5	202.1	−0.5	41.9	14.4	30.6	63.4
33	6/4/11	192.8	−31.7	190.5	−15.4	62.7	11.7	33.1	77.2
Mean	3	188.6	−0.4	187.7	−4.8			56.6	75.0
		k=7; α_{95}=49°		k=19; α_{95}=28.8°					
McFadden SCOS =		2.537		2.264 (95% critical value: 2.076)					

Note: No = samples demagnetized; N/n = samples/specimens used in site mean calculation; dec = declination; inc = inclination; VGP = virtual geomagnetic pole; k = best estimate of Fisher's precision parameter; α_{95}(°) = radius of the cone of 95% confidence. Data for the intrusive rocks are also shown corrected for the regional dip of the Todos Santos Formation at Cintalapa (15° to the northeast). The fold test statistics were calculated according to the isolated observation and the correlation tests of McFadden (1990). The test is inconclusive for the data from Motozintla and positive for the data from Matias Romero–Cintalapa.

†Sites 20 and 21 are not included in the final calculation.

*Indicates sites where linear segments and great circle trajectories were used.

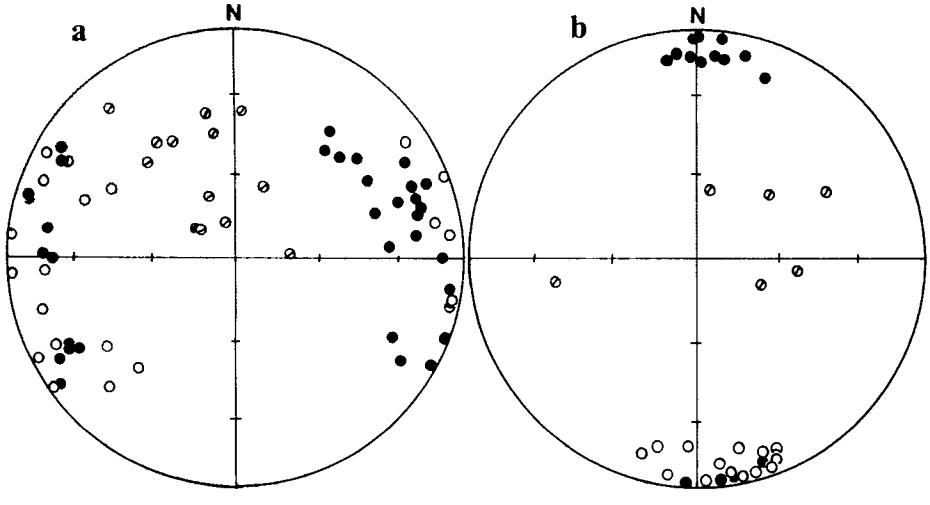


Figure 5. Equal area projections of the characteristic magnetizations isolated after demagnetization experiments (specimen directions are plotted). Open (closed) symbols are projected on the upper (lower) hemisphere, and crossed symbols are poles to great circle demagnetization trajectories. (a) Component C (Permian intrusives); (b) component B (Middle-Late Jurassic? dikes and granites).

tively simple magnetic behavior. Thermal and AF demagnetization isolated a well-defined westerly and shallow downwards magnetization, which unblocks at temperatures between 300 and 500 °C and inductions of 30 mT.

Typical demagnetization diagrams of the Todos Santos samples collected near Motozintla are shown in Figure 7. NRM intensities range between 0.24 and 15.9 mA/m. In sites 31 to 33, thermal demagnetization removes a low stability northerly and downwards component (component A above), isolating a southerly and shallow upwards or downwards component with maximum unblocking temperatures as high as 680 °C. The directions for the ChRM isolated in the Todos Santos Formation are plotted in Figure 8, both in *in situ* and tilt-corrected coordinates.

FIELD TESTS

Conglomerate Test

The clasts sampled in the uppermost Todos Santos Formation are well rounded and range in size from about 1 to 10 cm. The majority of the clasts are granitic, but a few volcanic clasts lithologically similar to the samples from sites 17 and 18 in Cintalapa were also observed. Typical clast demagnetization diagrams are plotted in Figures 9b and 9c, and NRM characteristic directions are plotted in a stereonet in Figure 9a. The clasts are characterized by a complex magnetization. Thermal and AF demagnetization remove an intermediate unblocking temperature (or coercivity) component that is often stable as high as 540 °C. Heating the samples at higher temperatures brings only small changes in the direction and intensity of the magnetization, and the end points of the magnetization vectors measured after each step form "clusters" on the stereonet and on the demagnetization diagram; then, they become magnetically unstable at higher temperatures, so those demagnetization steps are not plotted in Figure 9. The mean of the cluster of vector end points has been interpreted as the characteristic magnetization for each clast. The ChRM vectors are scattered in the equal area projection (Figure 9a), suggesting that this magnetization was acquired prior to deposition. In three samples, the component removed before reaching a stable end point is a southerly and shallow component similar to the B component (as in Fig. 9b). This possibly indicates that acquisition of component B postdates deposition of the conglomerate.

Fold Test

Using the data for the Todos Santos Formation sampled near Matias Romero and Cinta-

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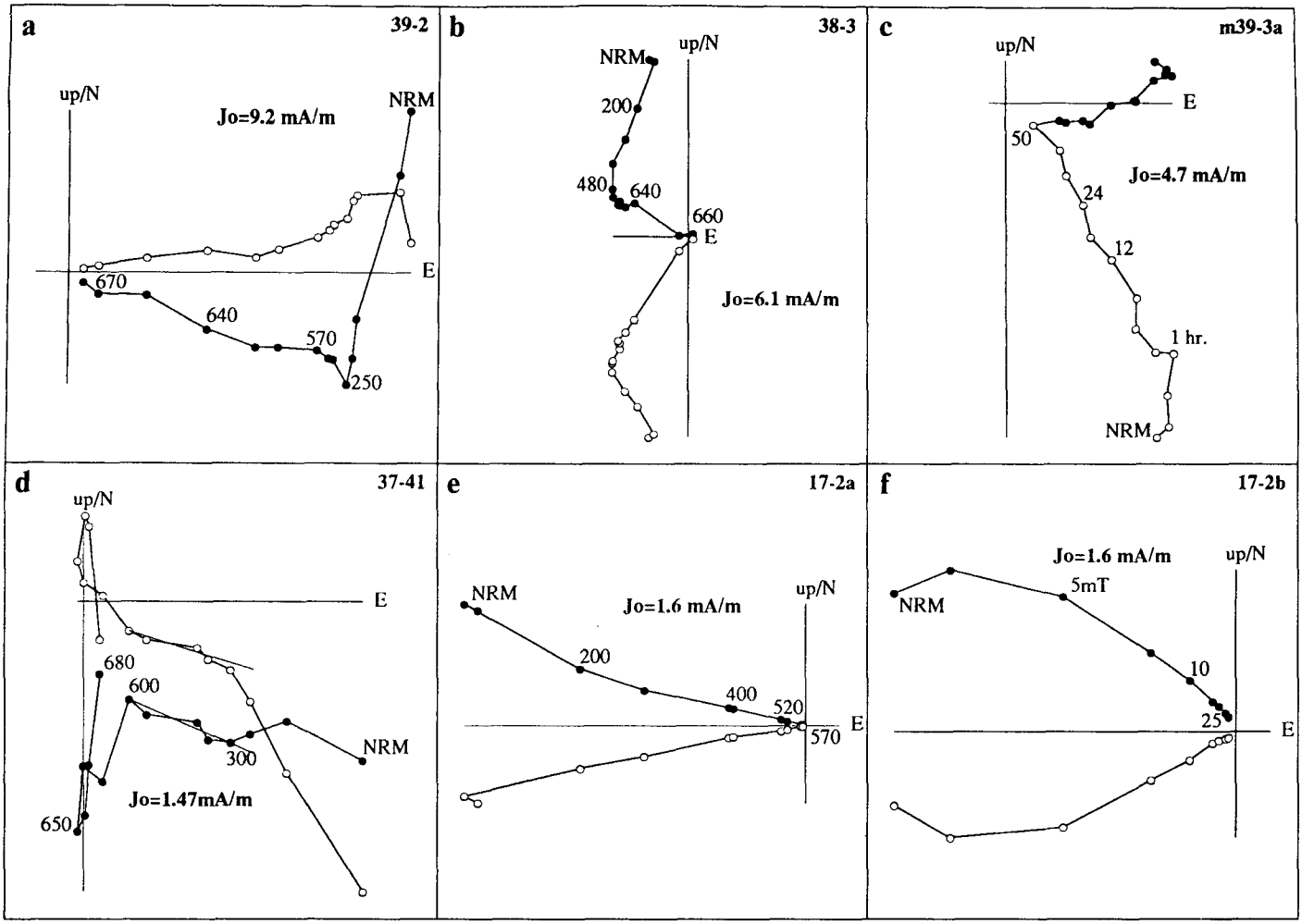


Figure 6 (a–f) Representative demagnetization diagrams for specimens of the northern sites collected in the Todos Santos Formation. Symbols are as in Figure 3. All samples are plotted in geographic coordinates. a, b, and d are thermal demagnetization diagrams, whereas c illustrates the results of chemical demagnetization. Sample 37-41 (d) shows three components of magnetization. After removing a steep component, the characteristic magnetization identified in other specimens unblocks between 300 and 600 °C. This is indicated by a line segment fitted through the points that define it. The third component is observed at higher temperatures (650 to 680 °C); this component is to the south and shallow, similar to component B of the intrusives. The samples from site 17 (e and f) were obtained from a volcanic unit within the lower portion of the Todos Santos Formation collected near Cintalapa.

lapa, a strongly positive regional fold test (k2/k1 = 3.96; Table 1; Fig. 9a) is obtained. The structural corrections for three of the sites are nearly the same and significantly different from the correction of the fourth site (site 38); thus, an appropriate fold test in this situation is the test on an isolated observation, or outlier, described by McFadden (1990). Results are presented in Table 1. The test indicates that the direction is pre- or synfolding (the maximum value for k is obtained after a partial unfolding of 78%). The mean directions after 78% and 100% unfolding, however, are not significantly different, and the improvement is not statistically significant. Thus, we regard the ChRM as pre-folding, that is, pre-Cretaceous.

The directions isolated in the Todos Santos Formation near Motozintla are all southerly, but scattered. Cluster improved only slightly after structural correction (*in situ*: k = 9, α_{95} = 49°; after tilt correction: k = 19, α_{95} = 28.8°; see Fig. 9b). Each site has a different structural correction; thus an appropriate fold test in this case would be the correlation test of McFadden (1990). The maximum value of k is obtained after a complete structural correction (100% unfolding). The null hypothesis of no correlation between the direction of magnetization and the direction and angle of dip, however, is rejected at the 95% confidence level, both in *in situ* coordinates and after tilt correction, resulting in a test that is inconclusive. This lack of resolution may be caused by undetected structural complexity or incomplete removal of a secondary component. Because the sites were collected in the vicinity of the CCP fault system, we believe that the scatter is more likely related to structural complexity (Carfantan, 1977; Anderson and others, 1985; Burkart and others, 1987).

INTERPRETATION

Intrusives

The ChRM of the Late Permian batholith (component C) is interpreted as a primary thermal remanent magnetization (TRM) acquired upon cooling. This interpretation is based on the

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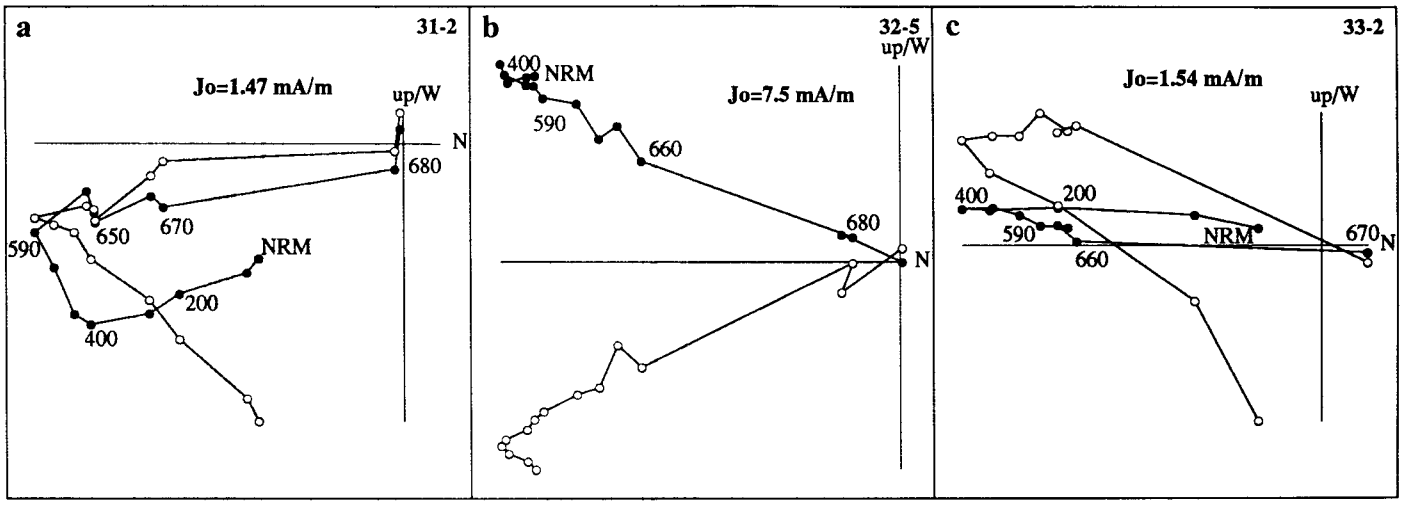


Figure 7. (a–c) Representative demagnetization diagrams from the southern sites collected in the Todos Santos Formation. Symbols are as in Figure 3. All samples are plotted in geographic coordinates. All three sites are characterized by a southerly and shallow, up or down, high-temperature, magnetization component carried by hematite.

observation of both normal and reverse polarities and the results of the conglomerate test. Moreover, a dike that crosscuts the granite has a different direction of magnetization (Fig. 4d), arguing against a regional remagnetizing event. The observation of normal and reverse polarities is consistent with the radiometric ages for the batholith, which indicate that the intrusions postdate the Kiaman superchron, which ended by 260 Ma (Molina-Garza and others, 1989). Within-site paleomagnetic directions in the Permian granites have somewhat large confidence intervals (Table 1); this is due in part to the small number of samples at some sites. The C component yields a mean paleomagnetic direction with dec = 261.3° and inc = –9.0° (N = 12 sites; k = 10; α_{95} = 15°; Table 1). The north-south component B is interpreted as a younger TRM or a partial TRM, presumably related to the Middle-Late Jurassic magmatic event proposed by Damon and others (1981). K-Ar ages for andesite dikes and granodiorites near Villa Flores, reported by Damon and others (1981), range in age between 141 ± 3 Ma and 174 ± 4 Ma, indicating the possible age range for the magnetization. The B component yields a mean paleomagnetic direction with dec = 176.5° and inc = –7° (N = 6 sites; k = 153; α_{95} = 5.4°; Table

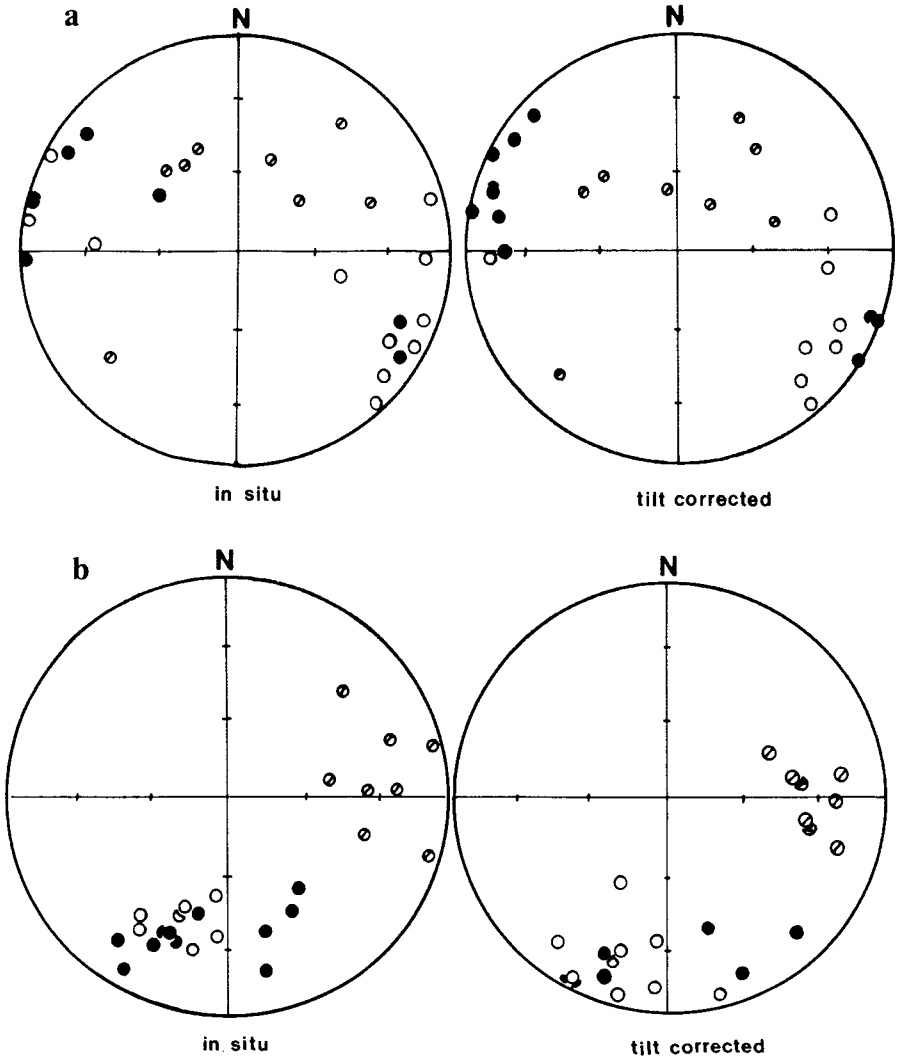


Figure 8. Equal area projections of the characteristic magnetization of the Todos Santos Formation (specimens). Symbols are as in Figure 5. (a) Northern sites (Matias Romero–Cintalapa); (b) southern sites (Motozintla).

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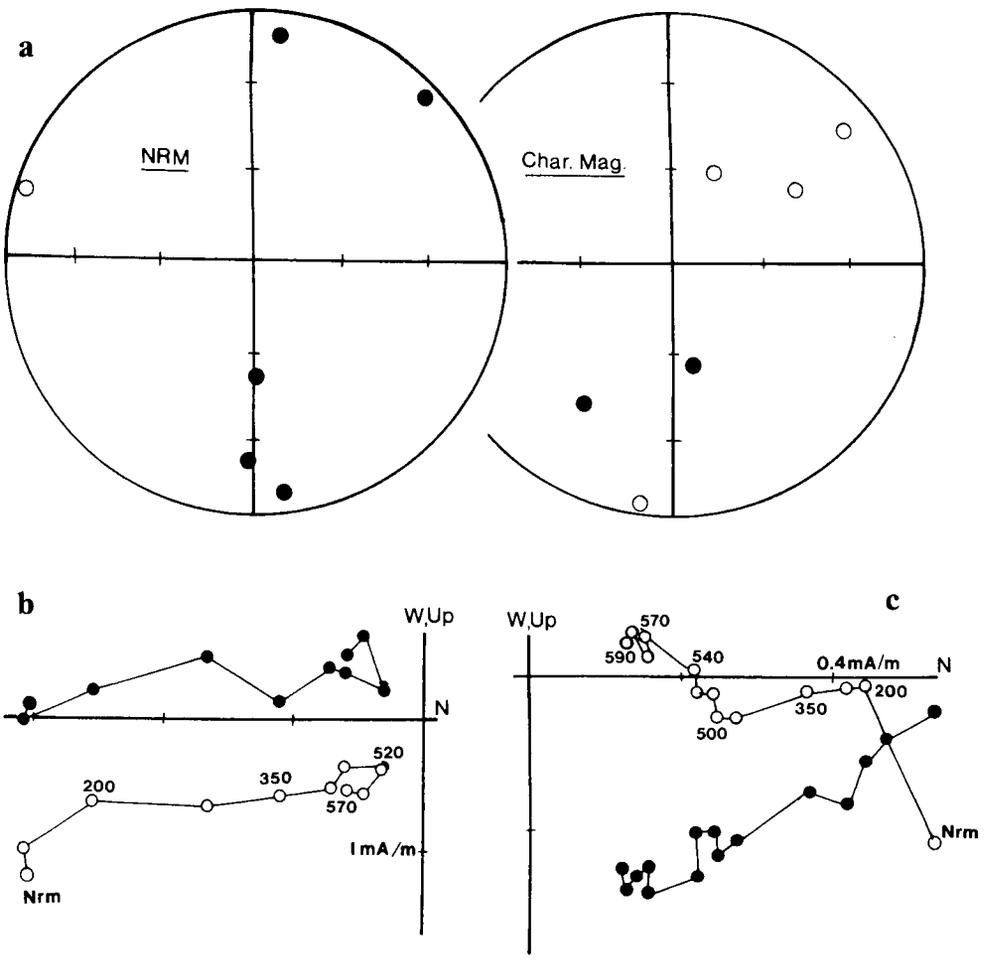


Figure 9. Demagnetization diagrams of selected samples and equal area projections of the paleomagnetic directions of granitic clasts sampled for a conglomerate test. Demagnetization diagrams are plotted as in Figure 3. (a) NRM directions and isolated end-point directions. (b and c) Demagnetization diagrams. Notice that in Figure 9b the component removed is southerly and shallow, similar to component B. This suggests that the B component post-dates deposition of the conglomerate.

1). The B and C components of magnetization were observed on both sides of the western extension of the CCP fault system (Fig. 1), suggesting that the batholithic complex has not been severely dislocated by recent motion along the fault.

Todos Santos

Based on a positive regional fold test and the presence of dual-polarity magnetizations, the characteristic component of the Todos Santos sequence near Matias Romero and Cintalapa is interpreted as a primary component acquired soon after deposition (Early to Middle Jurassic). This is further supported by the fact that some specimens of the Todos Santos sequence sampled near Matias Romero carry a southerly shallow overprinting component, similar to component B that was presumably acquired at later times. The sites of Todos Santos Formation (Matias Romero–Cintalapa) yield a mean paleomagnetic direction with dec = 293.7° and inc = 2.7° (N = 4 sites; k = 34; α_{95} = 15.8°; Table 1). The age of the magnetization of the Todos Santos Formation near Motozintla is more problematic. At this time, we favor the interpretation that the magnetization at this location was acquired in the Late Jurassic, according to the suggestion of Richards (1963) that the age of the Todos Santos Formation in this area is exclusively Late Jurassic. Nonetheless, the confidence interval of the mean direction is rather large. In view of the apparent problems in the structural interpretation of this area (Burkart and others, 1987; Anderson and others, 1985), results from this area will not be considered further.

DISCUSSION

Mean paleomagnetic poles were calculated using the virtual paleomagnetic pole (VGP) from each site. Paleomagnetic data are summarized in Table 2. The mean paleopoles are compared with the North American APWP in Figure 10. Observed paleomagnetic directions are compared with expected directions (calculated for Chiapas from the mean reference pole) in Table 2. Apparent tectonic rotation (R), apparent latitudinal displacement (D), and inclination flattening (F) (Beck, 1980), together with their associated errors (calculated according to Demarest [1983]) are also listed in Table 2.

The prevalent east-west directions of magnetization of the Chiapas batholith yield a paleopole (C) at 13.4°S, 176.1°E (A_{95} = 11.4°). The correlation of easterly (westerly) directions between sites that are widely separated argues against significant vertical axis rotation between the sampling sites; we will return to the issue of tectonic tilt below. Comparison with the Late Permian paleopole for stable North America (Molina-Garza and others, 1989; Van der Voo, 1990) indicates that western Chiapas has experienced a large rotation since Late Permian times (R = –78° or +102° ± 11.5°). By inference, this rotation is extrapolated to the entire Maya Block. The sense and amount of rotation cannot be determined from these data alone, since the low (near equatorial) latitudes make it difficult to determine whether the easterly directions correspond to magnetizations acquired during a period of normal or reverse polarity. The directions of the Todos Santos Formation (northern sites), however, yield a paleopole (TS) located at

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TABLE 2. SUMMARY OF PALEOMAGNETIC DATA FROM THE MAYA BLOCK

Rock Unit	B [n]	dec (°) (observed)	inc (°) (observed)	pole long °E	pole lat °N	K	A_{95}	dec (°) (expected)	inc (°) (expected)	A_{95}(°)	R(°)	±ΔR	F (°)	±ΔF	D (°)	±ΔD
Chiapas Massif*	12	261.3	-9.0	176.1	-13.4	15.4	11.4	339.0	-28.7	5.4	-77.7	11.5	-19.7	12.3	-10.6	8.9
Late Permian		263.0	2.5	179.0	-9.4			339.0	-28.7	5.4	-76.0	11.4	-31.2	12.3	-14.4	8.9
Todos Santos F.	4	293.7	2.7	170.6	22.6	56.8	12.3	357.0	-20.2	3.2	-63.3	11.4	-22.9	12.1	-1.9	9.0
Early-Mid-Jurassic								356.1	-1.7†	5.8	-62.4	11.9	-4.4	13.8	-2.4	9.6
B component	6	176.5	-7.0	97.4	77.4	269.2	4.1	176.6	3.8†	5.8	-0.1	5.6	-10.8	9.0	-5.5	5.0
Mid-Late Jurassic		177.2	0.9	97.6	73.3			176.6	3.8†	5.8	0.6	5.6	-2.9	9.0	-1.4	5.0
P. Hondo-Grup. F.	4	177.8	0.6	95.4	74.2	48.0	13.0	176.5	3.4†	5.8	1.3	10.1	-2.8	12.3	-2.2	10.1
Mid-Late Jurassic§																
Todos Santos F.	3	187.7	-4.8	56.6	75.0	23.7	25.9									
Late Jurassic?																
San Ricardo F.	[89]	340.3	19.9	160.0	69.8	25.4	3.0	340.4	20.7	9.4	-0.1	7.1	0.8	12.7	-1.8	7.0
latest Jurassic																

Note: B = sites (samples); dec = declination; inc = inclination. K and A_{95} are the estimates of the precision parameter and the cone of 95% confidence of the mean pole. Expected declinations were calculated using the North American reference pole from the data compiled by Van der Voo (1990) and recently published data. Flattening F and latitudinal displacement D are positive for northwards displacement, and apparent tectonic rotations (R) are positive for clockwise rotations.

*The directions of granites in Chiapas are given *in situ* (first) and corrected for the regional tilt of the Todos Santos near Cintalapa.

†Indicates the Middle Jurassic reference direction. The alternative reference direction given corresponds to the Early Jurassic for the Todos Santos and the Late Jurassic reference in the case of the B component, respectively.

§The magnetization of the Early Permian Paso Hondo and Grupera Formations has been reinterpreted as a Mid-Jurassic overprint.

22.6°N, 170.6°E, intermediate between the (rotated) C paleopole and the North American APWP (Fig. 10). Therefore, the westerly directions have been interpreted as normal polarities, and the rotation, as counterclockwise (R = -78°). The observed inclination of the ChRM (component C) of the Late Permian batholith (-9°) indicates a near equatorial paleolatitude for the sampling area. The observed inclination is significantly shallower than the inclination expected when assuming stability with respect to the craton (Table 2). We realize that paleomagnetic results from plutons may be used only with reservation to determine paleolatitudes, because of the inability to determine the paleohorizontal. A small regional eastwards tilt of the Chiapas batholith has been previously proposed by Mexican petroleum geologists to explain sedimentary and structural patterns in the Gulf coastal plain (Lopez-Ramos, 1979). Near Cintalapa, the Mesozoic sequence dips gently to the northeast, but farther south, along the Ocosocuautla–Villa Flores highway, nearly horizontal Cretaceous sediments overlie the granitic basement. If applied, a structural correction equal to that required to restore beds of the Todos Santos near Cintalapa to horizontal has negligible effect on the magnetization declination of the Permian granites and a small effect on the inclination (*in situ*: D = 261.3°, I = -9°; tilt corrected: D = 263°, I = +2.5°; Table 1). Therefore, our conclusions with respect to the rotation of the Maya Block are not affected whether tilting of the batholith is taken into account or not. As a result of the uncertainty on the tilt correction for the batholith, however, the paleolatitude determined for the Late Permian granites ranges between 5°S (*in situ*) and 1°N (if corrected for the tilt of the Todos Santos near Cintalapa). Similarly, the paleolatitude data indicate a relative post-Permian southward displacement of the Maya Block of 1,200 ± 900 km (*in situ*) or

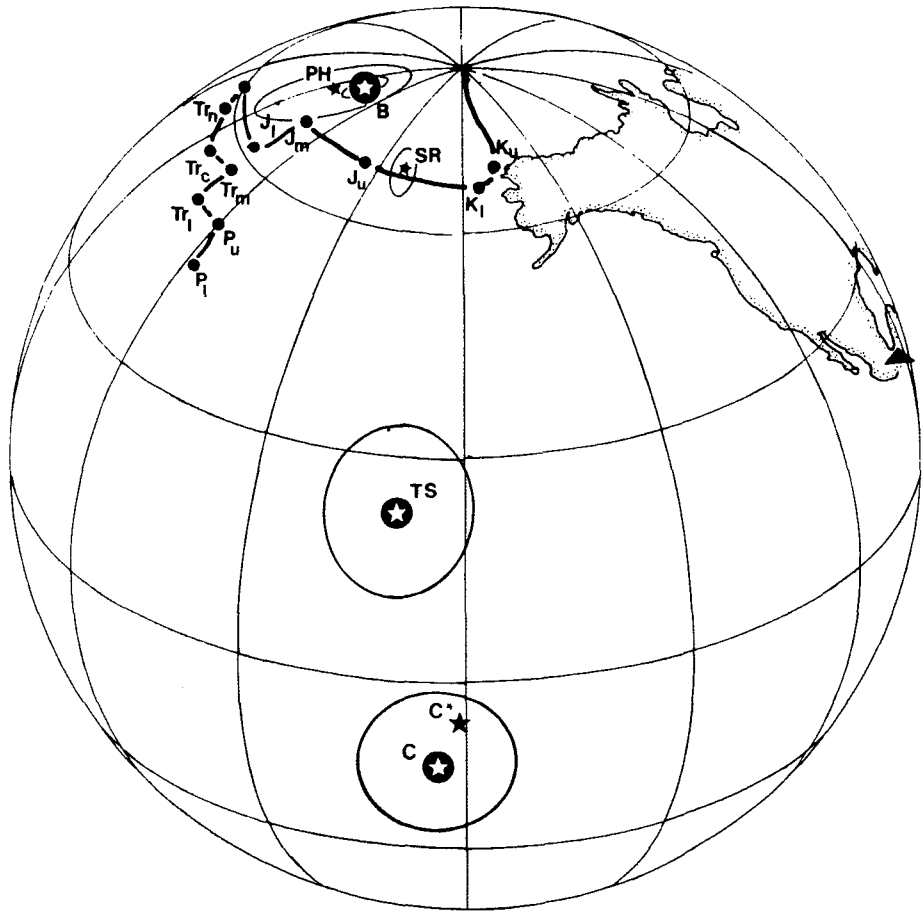


Figure 10. Paleomagnetic poles obtained in the present study (open stars) plotted in an orthographic globe projection (centered at 30°N, 180°E). C = Permian Chiapas Batholith; TS = Todos Santos Formation (northern sites); B = Jurassic granites and dikes. A closed star (C*) indicates the location of the Late Permian pole after a tilt correction similar to the trend of the Todos Santos Formation in Cintalapa. The North American APWP from the Permian to Tertiary times was calculated from data compiled by Van der Voo (1990) and data recently published, but not included there. The poles obtained by Gose and Sanchez-Barreda (1981) for the Early Permian Paso Hondo and Grupera Formations (PH), and by Guerrero and others (1990) for the latest Jurassic San Ricardo Formation (SR) have been plotted for reference. The small triangle indicates the sampling area.

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1,550 ± 900 km (tilt corrected) with respect to stable North America (Table 2).

The uncertainty on the age of the Todos Santos Formation has a very small effect on the estimate of the amount of rotation indicated by its ChRM. If this unit (and its magnetization) is Early Jurassic (post–earliest Jurassic cusp in the reference APWP) to Middle Jurassic, the amount of rotation is ~63° ± 11°. Such an uncertainty on the age, however, has a more significant effect on the estimate for latitudinal displacement. The paleolatitude determined for the Todos Santos Formation (4°N) indicates a relative southward displacement of 240 ± 960 km (assuming a Mid-Jurassic age, and using the Middle Jurassic reference pole), or 1,300 ± 900 km (for an Early Jurassic age).

The B component yields a paleopole (B) located at 77.4°N, 97.4°E (73.3°N, 97.6°E if corrected for the tilt of the Todos Santos in Cintalapa as discussed above) that plots near the North American Middle Jurassic reference pole (at high latitudes as suggested by recently published results from New England and the Newark basin [Van Fossen and Kent, 1990; Witte and Kent, 1990]). Late Jurassic and younger stability of the Maya Block with respect to North America has previously been postulated by Guerrero and others (1990) using paleomagnetic data from the Kimmeridgian-Portlandian San Ricardo Formation. If the interpretation of the age of component B is related to the magmatic event proposed by Damon and others (1981), the position of pole B (Middle-Upper Jurassic dikes and granites) indicates stability of the Maya Block since at least Oxfordian times. Pole B (Fig. 10) is also not significantly different from the pole obtained by Gose and Sanchez-Barreda (1981) for limestones of the Late Pennsylvanian–Early Permian Paso Hondo and Grupera Formations northeast of Motozintla (Table 2). Hence, it is logical to interpret this pole (PH in Fig. 10) as a Jurassic remagnetization.

PALEORECONSTRUCTIONS AND EVOLUTION OF THE GULF OF MEXICO

Our paleomagnetic data strongly favor those models for the evolution of the Gulf of Mexico that indicate large counterclockwise rotation of the Maya Block as the mechanism for the opening of the Gulf (White, 1980; Anderson and Schmidt, 1983; Pindell, 1985; Dunbar and Sawyer, 1987; Ross and Scotese, 1988). In these models, the position of the Maya Block in the

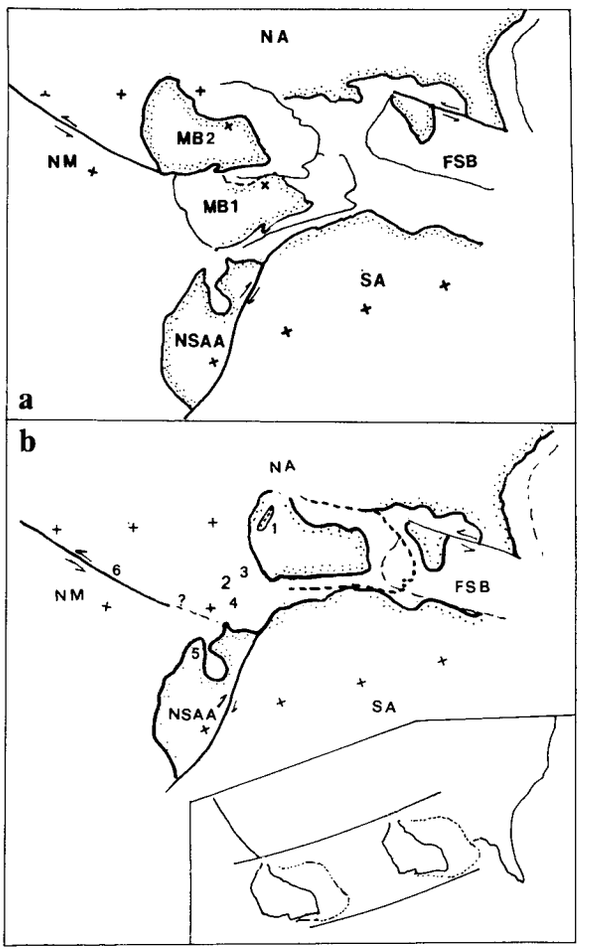


Figure 11. (a) Late Paleozoic–Early Mesozoic reconstruction of the Gulf of Mexico region, after Ross and Scotese (1988). The Maya Block is shown in the original position proposed by Ross and Scotese (MB1) and after an additional rotation of 20° (MB2), as indicated by the paleomagnetic data from Chiapas, using the same Euler pole of the original reconstruction. NA = North America; NM = Northern Mexico; SA = South America; NSAA = Northwestern South America allochthonous; FSB = Florida Straits Block. (b) Preferred position of the Maya Block. This reconstruction has as its basis the paleomagnetic data and the additional constraints of complete closure of the Gulf, the consideration of attenuation of the crust (Dunbar and Sawyer, 1987), restoration of post-Permian displacement of allochthonous terranes in northwestern South America (Pindell, 1985), and the reconstruction of a semi-continuous belt of Permo-Triassic calc-alkaline plutonism as explained in the text. (1) Chiapas batholith; (2) Tamaulipas batholith; (3) Burgos basin; (4) Tampico-Misantla basin; (5) Guajira peninsula; (6) Sonora-Mojave megashear. In the inset we illustrate alternative locations of the Maya Block, which are indistinguishable by paleomagnetic methods.

TABLE 3. EULER POLES AND ROTATION ANGLE FOR THE MAYA BLOCK IN SELECTED RECONSTRUCTIONS OF WESTERN EQUATORIAL PANGEA

Reconstruction parameters: Reference:	lat °N	long °E	angle (°)	Todos Santos pole TS °E/°N	Permian intrusives pole C °E/°N	Permian intrusives pole C* (t.c.) °E/°N
Dunbar and Sawyer (1986)	-25.00	101.00	45.0	128.4/54.1	154.6/23.5	156.4/28.1
Pindell (1985)	-29.74	99.60	43.0	129.8/52.0	153.0/20.4	154.9/25.0
Ross and Scotese (1988)	-27.95	92.10	53.3	116.3/60.9	147.0/30.6	149.0/35.2
Anderson and Schmidt (1983)	-29.90	86.30	58.0	109.2/64.9	142.3/34.4	144.9/38.9
Ross and Scotese (1988) +20°	-27.95	92.10	73.3*	73.8/61.9	128.2/43.3	128.7/48.2
This study	-24.50	95.30	75.5*	70.6/60.0	128.4/45.6	128.2/50.5
Reference pole:			Mid-Jurassic	98.8/71.8	L. Per. 120.3/52.2	120.3/52.2
			Early Jurassic	92.4/62.5		
			Tr-J cusp	60.0/60.0		

Note: poles from Chiapas have been rotated to North American coordinates and are compared with the reference pole. If the Maya Block was attached to North America during Late Permian to early Middle Jurassic times, the rotation should have brought poles from the Maya Block into coincidence with the craton's reference pole.

*Indicates rotation parameters used in Figure 11.

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interior of the Gulf between South and North America was suggested based on three main lines of evidence: (1) the assumption that salt provinces in the northern and the southern portions of the Gulf (the Louann salt and the Campeche salt basin) were originally adjacent, (2) the correlation of late Paleozoic sedimentary units in northern South America (Chandua and Sierra de Perija) with similarly aged units in Chiapas and the Maya Mountains (Anderson and Schmidt, 1983), and (3) geometrical constraints that required complete closing of the Gulf in Permo-Triassic times.

Using paleomagnetic data alone, it is very difficult to determine whether one reconstruction should be preferred over the other, because they differ only in minor detail and the paleomagnetic resolution is not sufficient. The model of Anderson and Schmidt (1983), however, is based on a relatively loose fit of the continents in the equatorial region (Bullard and others, 1965), which is not supported by paleomagnetic data (Van der Voo and others, 1976). Therefore, this reconstruction must be considered unsatisfactory. The pre-rift configurations of Pindell (1985) and Ross and Scotese (1988) incorporate total closure of the Gulf, use a substantially tighter fit of the continents, restore attenuation of the crust to its (assumed) original thickness (Dunbar and Sawyer, 1987), and retract, where possible, post-Permian displacements along intercontinental fault systems such as in northern South America and northern Mexico (Silver and Anderson, 1974). As we mentioned above, these models are practically indistinguishable by conventional paleomagnetic methods. The paleomagnetic data, however, indicate that in order to bring the poles from Chiapas into agreement with the North American reference, a greater rotation of the Maya Block is required than previously suggested. This is illustrated in Figure 11a.

The rotation parameters of the reconstructions discussed above, together with the paleomagnetic data from the Permian granites and the Todos Santos Formation (rotated into North American coordinates), are presented in Table 3. Using similar rotation poles as the models above, 15° to 25° of additional rotation of the Maya Block is required to bring the paleopoles from Chiapas (poles C and TS) into coincidence with the Late Permian and Early–Mid-Jurassic segments of the cratonic reference APWP (assuming, once again, that Chiapas has been part of the Maya Block since the Late Permian and that internal deformation within this block has not been important). Position 2 for the Maya Block in Figure 11a shows the effect of an additional 20° of rotation of the Maya Block based on the configuration of Ross and Scotese (1988). Note, however, that paleomagnetic data cannot detect paleolongitudes. We illustrate this lack of resolution by plotting the Maya Block in alternative (equal-latitude) positions in the inset of Figure 11a. Thus, other geological and geophysical considerations need to be included in order to constrain the precise location of the Maya Block.

Figure 11a shows that additional 20° rotation alone does not provide a satisfactory result. In fact, it causes the overlap of the Maya Block with portions of Texas generally considered "stable." A small adjustment to the rotation pole produces a more easterly position, which we show in Figure 11b (see reconstruction parameters in Table 3). The preferred reconstruction of Figure 11b accomplishes two important results, which we discuss below.

First, the reconstruction of Figure 11b changes the trend of the Chiapas batholith from northwest to (paleo) north-northeast. The latter orientation results, potentially, in a better alignment of a belt of Permo-Triassic calc-alkaline magmatism extending from the northern portion of the Gulf of Mexico, into the Chiapas batholith, continuing southwards into the Tamaulipas batholith (Lopez-Ramos, 1979), and possibly including Late Paleozoic intrusions in the Burgos and the Tampico-Misantla basins, in northeast Mexico (Lopez-Ramos, 1979). This belt of late Paleozoic magmatism could possibly extend farther south. Late Paleozoic intrusions exist in Jalapa (eastern portion of the Trans-Mexican volcanic belt), Oaxaca (southern Mexico), and, finally, the Central Cordillera and Paraguana Peninsula in northern Colombia (Irving, 1975; Pindell, 1985). This belt of magmatism possibly represents the remnants of an arc or Andean margin that existed between Gondwana and North America in Late Paleozoic times.

Second, the reconstruction of the Maya Block in Figure 11b leaves a gap in the eastern portion of the Gulf. Rowley and Pindell (1989) have pointed out that a tight closure in the equatorial Atlantic compatible with opening flow lines requires the fitting of the Demarara rise and the Guinea plateau. This can in turn be achieved by rotating the Florida Straits Block to a more westerly position. The reconstruction of Figure 11 is consistent with that interpretation, regardless of whether such a configuration is attained by closing the south Georgia rift (Klitgord and others, 1983) or by motion along the south Florida Elbow fault (Rowley and Pindell, 1989).

We propose that rotation of the Maya Block consists of two components: (1) separation of South America and the Maya Block from North America, coeval with opening of the central Atlantic, and (2) independent separation (and rotation) of the Maya Block from South America in late Middle Jurassic times. Thus, the Maya Block behaved briefly as a true microplate before arriving at its present position in the Middle or early Late Jurassic.

Thick deposits of red beds east of the Tamaulipas batholith suggest that the eastern flank is a large fault that was active in Late Triassic–Early Jurassic times (Lopez-Ramos, 1979). We believe that rotation of the Maya Block was probably accommodated along this fault (the Tamaulipas-Chiapas fault of Pindell [1985]). The Maya Block and other terranes in southern Mexico (for example, Oaxaca and Acatlan) were apparently in (minor?) relative motion in the Middle Jurassic (Fang and others, 1989). Thus, the consolidation of southern Mexico probably did not occur until Late Jurassic times.

Our conclusions with respect to the rotation of the Maya Block are, once again, based on the extrapolation of paleomagnetic data from Chiapas to the entire Maya Block. Further paleomagnetic data from other areas of the Maya Block would be desirable in order to demonstrate the absence of significant internal deformation within the Maya Block and tectonic coherence between the peninsula, Chiapas, and other areas of crystalline basement identified within the Block.

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